BIOLOGY SSC-I — Class 9

Chapter 6

MOLECULAR BIOLOGY

Detailed Study Notes

Based on BISE Islamabad Textbook | Pages 78–92

SLOs (After completing this chapter, the student will be able to:)	
SLO 1	Define Biochemistry / Molecular Biology
SLO 2	Outline the various types of common biomolecules (DNA, RNA, Proteins, Lipids, Carbohydrates) including their locations in the cell and main roles
SLO 3	Outline the structure and function and sources of proteins with structure of amino acids
SLO 4	Outline the structure, function and sources of lipids
SLO 5	Define Carbohydrates and Outline the structure, function and sources of Carbohydrates
SLO 6	Identify carbohydrates as monosaccharides, disaccharides and polysaccharides
SLO 7	Describe briefly the structure of DNA as a double helix macromolecule made of nucleotides with base pairing in between the two helices through complementary base pairing
SLO 8	Outline function of DNA as carrier of hereditary information
SLO 9	Describe briefly the structure of RNA as single stranded macromolecule made of nucleotides with nitrogenous base overhangs
SLO 10	Outline the function of RNA as aid in converting hereditary information into useful proteins
SLO 11	Outline how information in the DNA is converted to information on RNA and then into proteins

6.1 Biochemistry / Molecular Biology

Definition: Biochemistry is the study of different chemical compounds and the chemical processes taking place within the living organisms.

Relation: Biochemistry is related to both **biology** and **chemistry**. Therefore, biochemistry solves the problems faced by living organisms by using the knowledge and techniques of chemistry.

Molecular Biology vs Biochemistry: Sometimes biochemistry and molecular biology are taken as same but **molecular biology is the specialized branch of biochemistry** that is mainly concerned with the interaction of bio-molecules within the cells and biochemical processes like DNA replication, transcription and translation.

6.1.1 Importance of Biochemistry

Study of biochemistry is greatly helpful to explore the **cell biology and anatomy** as all the structures of the living organisms (cells, tissues and organs etc.) have specific biochemical organization. Similarly, knowledge of biochemistry is essential to study the **physiology of organisms** because life processes such as photosynthesis, respiration, digestion, inheritance are explained in biochemical terms.

6.1.2 Bioelements

Bodies of living organisms are composed of protoplasm, which is made of certain elements called **bioelements**. Out of **92** naturally occurring elements of earth, **16** of these are bioelements. On the basis of their proportions in the living organisms, bioelements are further classified into **Major** and **Minor** bioelements.

Major Bioelements: Six most common bioelements that constitute **99% of protoplasm**, so called major bioelements.

Element	Symbol	% in Human Body	Role
Oxygen	O	65%	Most abundant; component of water and biomolecules
Carbon	C	18%	Backbone of all organic molecules
Hydrogen	H	10%	Component of water and carbohydrates
Nitrogen	N	3%	Component of proteins and nucleic acids
Calcium	Ca	2%	Bones, teeth, muscle contraction
Phosphorus	P	1%	ATP, DNA/RNA backbone, bones

Minor Bioelements: Remaining ten bioelements constitute only 1% of the biomolecules — so called **Minor** bioelements:

Potassium (K), Sulphur (S), Chlorine (Cl), Sodium (Na), Magnesium (Mg), Iron (Fe), Copper (Cu), Manganese (Mn), Zinc (Zn), Iodine (I)

★ **Note:** Compare the ratios of bio-elements (important short answer question): Major = 6 elements = 99% | Minor = 10 elements = 1%

6.2 Biological Molecules (Biomolecules) — Table 6.1

Definition: Molecules produced and utilized in the metabolic processes within the living organisms are called **biological molecules**. Bio-molecules are the building blocks of life and perform important functions in living organisms like growth, development, energy production, reproduction.

Classification: Biomolecules may be **inorganic** (water and salts) and **organic** (carbohydrates, lipids, proteins and nucleic acids). Complex organic biomolecules are **polymers** of their building blocks called **monomers**. Major organic bio-molecules, their locations inside the cell and main roles are given in Table 6.1:

Biomolecule	Examples	Location in Cell	Main Roles
Carbohydrates	e.g. Glucose, Sucrose, Starch, Cellulose, Chitin	Cell wall, cell membrane, Smooth ER, Golgi bodies	Used as food, energy source, components of plant and fungi cell

			walls, cell surface markers, component of DNA and RNA as 5-carbon sugar
Proteins	e.g. Haemoglobin, Actin, Myosin, Hormones, Enzymes, Antibodies	Cytoplasm, Rough ER, Golgi bodies	Energy source, component of cell membrane, gaseous transport, speed up chemical reactions (enzymes), immunity, component of chromosomes
Lipids	e.g. Fats, Phospholipids, Waxes, Oils, Vitamin D	Cell membrane, Smooth ER	Provide cells with long-term energy, component of cell membranes
DNA	Component of chromosomes (genes)	Nucleus (Chromosomes)	Component of chromosomes; Store and pass on genetic information
RNA	mRNA, tRNA, rRNA	Nucleolus, Cytoplasm	Tools for translation, protein synthesis

6.3 Proteins

Proteins are main **structural and functional components of the cell**, which make up around **55% of the dry weight of cell**. They are present in all types of cells and contribute in the variety of cell functioning. Proteins are synthesized in the cells by **ribosomes**.

6.3.1 Chemical Composition of Proteins

Proteins molecules mainly composed of **C, H, O, N** and sometimes **P, S, Fe** etc. Chemically proteins are **polypeptide chains** which are **polymers of amino acids**.

6.3.2 Amino Acids

Definition: Amino acids are the **building blocks or monomers of proteins**. There are about **20–25 different types** of amino acids commonly found in proteins.

Structure: Each amino acid contains a central carbon atom called **α (alpha) carbon**. Four different groups are attached to α-carbon:

- A hydrogen atom (–H)
- An amino group ($–NH_2$)
- A carboxyl group (–COOH)
- An alkyl group labelled –R. Each amino acid has a

R-group: Each amino acid has a different –R group which determines its chemical properties.

Peptide Bond: Bond between two amino acids is called a **peptide bond**. Number, type and sequence of amino acids in each polypeptide chain is specific.

■ Diagram: General Structure of an Amino Acid (Fig. 6.2)

```
                    H
                    |
          H₂N  ——  C  ——  COOH
                    |
                    R  (variable side chain)

   Four groups attached to alpha carbon:
   1. –H       (Hydrogen atom)
   2. –NH₂     (Amino group)
   3. –COOH    (Carboxyl group)
   4. –R       (Alkyl group — different for each amino acid)

   Peptide bond: –COOH of one amino acid bonds
   with –NH₂ of next amino acid
```

6.3.3 Structure of Proteins (4 Levels)

The structure of proteins is classified as **primary, secondary, tertiary** and in some cases **quaternary**. These structures are based on the level of complexity of the folding of a polypeptide chain.

Level	Description	Key Feature
Primary	A linear polypeptide with a specific number, type and sequence of amino acids is called primary structure.	Linear chain — sequence determines all higher structures
Secondary	After synthesis, polypeptide folds upon itself due to additional bonds to form higher structural level (secondary). Polypeptide folds to form alpha-helices or beta-pleated sheets.	Alpha-helix / Beta-pleated sheet
Tertiary	Further folding of secondary structure with even extra bonds (e.g. disulphide bonds) resulting in fibrous or globular shape.	3D globular or fibrous shape
Quaternary	Two or more polypeptides bonded with each other. Shape of the proteins also depends upon structural level in the protein.	Multiple polypeptides e.g. Haemoglobin

★ **Note:** After synthesis a protein does not remain in its primary structure. Polypeptide folds upon itself due to additional bonds to form some higher structural level (secondary, tertiary or quaternary) to perform its specific role. Shape of the proteins (fibrous or globular) also depends on the structural level.

6.3.4 Functions of Proteins (Table 6.2)

Proteins are very important molecules in our cells. Each protein within the body has a specific function. They are involved in the formation of many structures of body. They also play many important roles with in body. **One gram of protein has 4 kcal of energy**.

Protein	Example	Function
Cell membrane protein	Many proteins in plasma membrane act as channel and carrier proteins. Some proteins at surface of cell act as antigen (define blood groups).	Act as channel and carrier proteins
Enzymes	Most of the enzymes are protein which speed up biochemical reactions.	Speed up biochemical reactions
Hormones	Protein hormones act as chemical messengers and regulate different functions in body e.g. growth, reproduction, digestion, glucose level etc.	Chemical messengers
Antibodies	Antibody proteins produced by WBCs.	Provide immunity against germs
Haemoglobin	RBCs have red pigment haemoglobin protein.	Transport of oxygen and carbon dioxide via blood
Fibrinogen	Blood plasma contains a blood clotting protein.	Stop bleeding at injuries
Actin and Myosin	Present in muscle cells.	Contraction and relaxation of muscles during movement
Collagen	Major component in structure of skin, bones, cartilage, muscles, tendons and ligaments.	Provides support and strength
Keratin	Component of hair, nails, feathers, horns and beaks.	Make them hard and protective

Histone	Attached to DNA.	To form chromosome

6.3.5 Sources of Proteins + Biuret Test (STEAM Activity 6.1)

Plant Sources: Plant seeds are most common source of proteins like **beans, lentils, peas, nuts**.

Animal Sources: Animal products are sources of high protein: **Milk, beef, mutton, poultry meat, egg, fish, seafood**.

Other Sources: Some proteins are obtained from **mushrooms**.

Experiment (Biuret Test)	Observation	Inference
Take 2 ml of test solution (protein solution), add 2 ml of concentrated solution of NaOH and two to three drops of 1% copper sulphate solution and then shake well.	Pink or Violet coloration is produced in test tube.	Protein is present in the solution.

6.4 Lipids

Definition: A group of different types of organic compounds which are all **hydrophobic in nature**, collectively called **lipids**. Hence, **lipids are a heterogeneous group of organic compounds which are insoluble in water** but soluble in organic solvent such as alcohol, ether etc. Lipids found in living organisms include **fats, oils, waxes and steroids**.

Composition: Molecules of lipids are composed of **C, H and O** like carbohydrates. However **one gram of lipids contain double amount of energy than carbohydrates** because they have relatively more carbon-hydrogen ratio than carbohydrates.

6.4.1 Structure of Lipids

Acyl Glycerides: Lipids which are mainly made up of **fatty acids and glycerol (an Alcohol)** are called **Acyl glycerides**. Fatty acids may be **saturated** or **unsaturated**.

Type	Description
Saturated Fatty Acids	Have all C–C single bonds. Are solid at room temperature e.g. cheese, butter, ghee etc.
Unsaturated Fatty Acids	Have double or triple C–C bonds and are liquid at room temperature e.g. soybean oil, mustard oil, olive oil etc.

Tri-acyl Glycerides: Most common types of acyl glycerides are **tri-acyl glycerides** which contains **one glycerol and three fatty acids**.

Glycerol + 3 Fatty Acids → Tri-acyl Glyceride

Phospholipids: Some lipids are even more complex and have **phosphate, nitrogen base along with glycerol and fatty acids** e.g. phospholipids and waxes.

Steroids: Some lipids may not have fatty acids in their formula rather they have **three ring structure** called **steroids** e.g. **cholesterol, vitamin D**.

■ Diagram: Structure of Different Types of Lipids (Fig. 6.6)

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1. TRIGLYCERIDE (Tri-acyl Glyceride):
   [Glycerol] — [Fatty Acid 1]
              └— [Fatty Acid 2]
              └— [Fatty Acid 3]

2. PHOSPHOLIPID (Cell Membrane Bilayer):
   HEAD: Glycerol + Phosphate + Nitrogen base
         (hydrophilic — water-loving)
   TAIL: 2 Fatty acid chains
         (hydrophobic — water-fearing)
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   3. STEROID (e.g. Cholesterol):
      Structure: 4 fused carbon rings (no fatty acids)
      Examples: Cholesterol, Vitamin D
```

6.4.2 Functions of Lipids

- **Phospholipids and cholesterol are components of** plasma membrane structure.
- **Lipids act as** energy stores in fat cells, liver and in blood. One gram of lipids provides 9.1 kcal of energy which is double than carbohydrates or proteins.
- **Many lipids act as important** steroid hormones e.g. testosterone, Aldosterone etc.
- **Some lipids act as** insulators against atmospheric heat and cold.
- **Waxes are used by insects for** constructing their hives.
- **Waxy cuticle of leaves acts as** protective and waterproof layer to avoid water loss.
- **Some vitamins are lipid in nature e.g.** vitamin A, D, E and K. Vitamin D is derived from cholesterol.

6.4.3 Sources of Lipids + Biochemical Tests (STEAM Activity 6.2)

Animal Sources: Foods containing large amounts of saturated fat include **red meat, butter, cheese, ghee, fish liver oil** etc.

Plant Sources: Some oils are obtained from plant seeds like **sunflower oil, soybean oil, peanut oil, coconut and palm oil**. These are also rich in unsaturated fats.

Insect Sources: Wax is also obtained from comb of many insects.

Test	Experiment	Observation	Inference
A. Spot Test	Put a drop of test solution (lipid solution) on a filter paper and allow it to dry.	A clear or translucent greasy spot appears on paper.	Lipid is present in the solution.
B. Insolubility Test	Take a small amount of fat (butter or ghee) in test tube and half fill with distilled water.	Fat and water remain separate.	Fat is insoluble in water.
C. Heat Test	Heat the water and fat in the test tube.	Fat melts on heating but don't mix water and make clear upper layer over the lower water layer.	Fat is insoluble in water and don't mix even in melted form.
D. Emulsion Test	Take 2 ml of the test material (lipid), add 2 ml of absolute alcohol and shake well. Now add 4 ml of cold water and shake well again then allow to stand.	A cloudy white suspension (emulsion) is formed after shaking.	Lipid is confirmed.

6.5 Carbohydrates

Definition: Carbohydrates are the important biological molecules present in all parts of cell. Carbohydrates are also called as **sugars or saccharides** due to their sweet taste. Word saccharide is derived from Greek word *"Sakcharon"* meaning sugar.

Composition: Carbohydrates are the compounds of **carbon, hydrogen and oxygen** with a ratio of **1:2:1** respectively and literally called **"hydrates of carbon"**.

Production: Most common and simple carbohydrate, **glucose**, is naturally produced in the green plants by the process of **photosynthesis** using CO_2 from air and H_2O from soil.

Energy: Plants use sunlight energy for food synthesis. This energy is stored as chemical energy in the bonds of glucose molecules and **4 kcal per gram of energy** is released when glucose is broken.

■ Diagram: Photosynthesis — Glucose Production

```
                    Light
```

$6CO_2$	+	$6H_2O$	——————→	$C_6H_{12}O_6$	+	$6O_2$
Carbon Dioxide		Water	Chlorophyll	Glucose		Oxygen

6.6 Classification of Carbohydrates

Carbohydrates are classified on the basis of **number of monomers** i.e. saccharide units in the molecule. Carbohydrates are generally classified into three groups:

6.6.1 Monosaccharides

Monosaccharides are the **simplest form of carbohydrates** which are primarily produced during photosynthesis. They are **white crystalline solids** and are **soluble in water**. They are sweet in taste. Number of carbon atoms in different monosaccharides may range from **3 to 7**.

Naming: Monosaccharides are named on the basis of number of carbon atoms in their structure:

Name	Carbon Atoms	Example
Trioses	3C	Glyceraldehyde
Tetroses	4C	Erythrose
Pentoses	5C	Ribose ($C_5H_{10}O_5$) — found in nucleotides of nucleic acids
Hexoses	6C	Glucose ($C_6H_{12}O_6$) — familiar hexose

Most Common: Most common monosaccharides are **pentoses and hexoses**. Monosaccharides are **not further hydrolysed**. If monosaccharides (mostly pentoses and hexoses) are in crystalline form they are found as **open chain structure** but when dissolved in water they are converted into **ring structure**.

■ Diagram: Open Chain and Ring Structure of Ribose (Fig. 6.10)

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Open Chain (Ribose):          Ring Structure:
   1CHO                       1CH₂OH
H  2C  OH                H    2C   H
H  3C  OH            HO   3C        4C  OH
H  4C  OH                   5C   O
   5CH₂OH                    H   OH

Ribose = C₅H₁₀O₅  (Pentose)   — present in RNA
Deoxyribose = C₅H₁₀O₄        — present in DNA
```

6.6.2 Disaccharides

Carbohydrates which release **two monosaccharides on their hydrolysis** are called **disaccharides** because disaccharide are formed due to bonding of two monosaccharides. They are also white crystalline solids. They are comparatively less soluble in water and are less sweet in taste. The molecular formula of disaccharide is $\mathbf{C_{12}H_{22}O_{11}}$. Examples of disaccharides are **sucrose, maltose and lactose**.

Disaccharide	Formed From	Equation	Source / Note
Sucrose	Glucose + Fructose	$C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$	Most familiar disaccharide; commonly used sweetener; known as cane sugar
Maltose	Glucose + Glucose	$C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$	Produced during hydrolysis of starch and glycogen; slightly sweet; very soluble in water
Lactose	Galactose + Glucose	$C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$	Naturally found in milk and milk products; the only common sugar obtained from animals

★ **Note:** Sucrose is the most familiar disaccharide, commonly used sweetener and known as cane sugar. The sucrose is formed by the condensation of glucose and fructose releasing a water molecule. Lactose is the only common sugar obtained from animals. Milk of all mammals is composed of 2–8% lactose.

6.6.3 Polysaccharides

Polysaccharides is largest group of carbohydrates. They are complex carbohydrates and **polymers of more than ten monosaccharides**. They are produced in the cells of plants and animals to **store extra carbohydrate food and energy** e.g., starch, glycogen. They are **not sweet in taste** rather usually tasteless. They yield many monosaccharide units upon complete hydrolysis for energy use. They are also **structural components of cell walls** in the cells of plant (cellulose), fungi (chitin) and bacteria (peptidoglycan).

Polysaccharide	Details
Starch	Storage polysaccharide in PLANTS. Present in cereals, wheat, barley, rice, maize, potato, sweet potato etc.
Glycogen	Storage polysaccharide in ANIMALS to store extra carbohydrate. Intermediate disaccharide during hydrolysis is maltose.
Cellulose	Structural polysaccharide — component of plant cell walls. Cellulose is obtained from plants e.g. Cotton is pure cellulose. Cellulose is present in whole grains.
Chitin	Structural polysaccharide — component of cell walls of fungi and exoskeleton of insects/arthropods.
Peptidoglycan	Structural polysaccharide — component of cell walls of bacteria.

Biochemical Tests for Carbohydrates (STEAM Activity 6.3)

Experiment	Procedure	Observation	Inference
Iodine test for Starch in solution	Take 5 ml water + small amount starch, boil. Add few drops of iodine solution into the clear starch solution.	Dark blue-black coloration is produced in test tube.	Starch is confirmed in solution.
Iodine test for Starch in food	Take food material like potato, cut a small slice and add a few drops of iodine solution on the potato slice.	Dark blue-black coloration is produced on potato slice.	Starch is confirmed in food.
Iodine test for Glycogen	Take glycogen solution in test tube and add few drops of iodine solution.	Red coloration is produced in test tube.	Glycogen is present in solution.
Iodine test for Cellulose	Take cellulose solution in test tube and add few drops of iodine solution.	No change in colour of iodine solution in test tube.	Starch and glycogen absent and Cellulose may be present.
Test for Glucose (Benedict's)	Take 5 ml of Benedict's solution, add a few drops of the test solution (glucose solution) and boil for 2–3 minutes.	A dirty green, yellow or red precipitates are formed.	Glucose is present.
Test for Glucose (Fehling's)	Take 3 ml of Fehling's solution in a test tube, add 3 ml of test solution (glucose solution) and heat.	Red precipitates are formed.	Glucose is present.

6.7 Nucleic Acids

Definition: Nucleic acids are the most important organic bio-molecules which **store and transfer hereditary information** from one generation to next. There are two main types of nucleic acid:

- Deoxyribonucleic Acid (DNA)
- Ribonucleic Acid (RNA)

Both types of nucleic acids (DNA and RNA) are made up of many units called **nucleotides**.

> ★ **Note:** Science titbits: Swiss physician Friedrich Miescher isolated a whitish material from nucleus of human pus cells. He named this material as nuclein as it was isolated from nucleus. Later on, nuclein was called nucleic acid on the basis of its acidic properties.

6.7.1 Structure of Nucleotide

Each nucleotide is made up of three components:

- **i.** Pentose (5C Sugar)
- **ii.** Nitrogen containing ring structure called
  Nitrogen base — attached at the **first carbon** of pentose sugar
- **iii.** Phosphate group — Acid attached with
  fifth carbon of pentose sugar

Pentose sugar in nucleic acid is of further two types:

- a) Deoxyribose — in DNA
- b) Ribose — in RNA

The nitrogen bases in nucleic acid are also of two main types:

Category	Nitrogen Bases
Purines (double ring structures)	Adenine (A) and Guanine (G)
Pyrimidines (single ring structures)	Cytosine (C), Thymine (T) and Uracil (U)

DNA nucleotides: The DNA molecule consists of four different types of nucleotides called **deoxyribonucleotides**. Each deoxyribonucleotide can have any one of the nitrogen base from **Adenine, Cytosine, Guanine and Thymine**.

RNA nucleotides: The RNA molecule also consists of four types of nucleotides called **ribonucleotides**. Each ribonucleotide can have any one of the nitrogen base from **Adenine, Cytosine, Guanine and Uracil**. So **DNA nucleotides can have Thymine while RNA nucleotides can have Uracil** as nitrogen base.

■ Diagram: Structure of a Nucleotide (Fig. 6.11)

```
            O
            ‖   Phosphate Group  (attached at C5)
        HO—P—O
            |
            OH   (1)
            |
      [———————————————————————]
      | Pentose Sugar (2)                 |
      | Ribose (RNA) / Deoxyribose (DNA)  |
      [———————————————————————]
            |
      Nitrogenous Base (3) — attached at C1
      (Adenine / Guanine / Cytosine / Thymine / Uracil)

    Fig. 6.12: Types of Nitrogen bases
    Pyrimidine (Single ringed): Uracil, Thymine, Cytosine, Adenine
    Purine     (Double ringed): Adenine, Guanine
```

6.7.2 Structure of DNA — Watson and Crick Model

DNA structure: DNA molecule is a **polynucleotide** as it is made up of many deoxyribonucleotides (Fig. 6.13). Each next nucleotide binds through its phosphate at the 3rd carbon of pentose sugar of previous nucleotide. Structure of

DNA was explained by two scientists, **James Watson and Francis Crick**. The main points of Watson and Crick Model of DNA are:

1.	The DNA molecule is a double stranded like a ladder and both strands are coiled around each other like a helix.
2.	Both strands of DNA molecule are antiparallel (oppositely oriented to each other).
3.	The back bone (uprights) of molecule are composed of sugar and phosphate part of nucleotide while bases form the rungs.
4.	The amount of adenine and thymine is always equal and amount of cytosine and guanine is always equal in DNA molecule.
5.	Base pairing between the both strands is highly specific. Adenine of one strand always pairs with thymine of other strand while cytosine of one strand always pairs with guanine of other strand. Due to this base pairing principle, the sequence of nucleotides in one strand is complimentary to the sequence of nucleotides in the other strand.
6.	Both strands of DNA are held by weak hydrogen bonds. There are two hydrogen bonds between adenine and thymine and three hydrogen bonds between cytosine and guanine.
7.	The diameter of double helix of DNA is thoroughly uniform and is about 2 nm.

■ Diagram: Watson & Crick DNA Double Helix (Fig. 6.13)

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5'━━━
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During sexual reproduction, gametes or spores are produced by **meiosis**. Male and female gamete (sperm and egg) contain **chromosomes (set of genes)** of parent generation. At fertilization, male and female gametes fuse to form **zygote**, thus genes of both parents are transferred to zygote. Zygote develops into the offsprings which are the next generation. Thus, genes are transferred from one generation to next though gametes or spores. So therefore, **DNA (genes) in the chromosomes carry the hereditary information from one generation to next**.

6.9 Structure of Ribonucleic Acid (RNA)

The RNA is a **polynucleotide helical chain** like DNA but it is **single stranded** nucleic acid. (Fig.6.13) In RNA **pentose sugar ribose** is present instead of deoxyribose. RNA molecule also differs from DNA in having **uracil nucleotide instead of thymine** nucleotide. Nitrogen bases of RNA molecule are **overhung** because they are not bonded to nitrogen bases of other (complementary) strand like DNA. Sequence of nucleotides in RNA is synthesized by using the **sequence of nucleotides in DNA by the process of transcription**.

Feature	DNA	RNA
Full Name	Deoxyribonucleic Acid	Ribonucleic Acid
Strands	Double stranded	Single stranded
Sugar	Deoxyribose	Ribose
Nitrogen Bases	A, T, G, C (Thymine)	A, U, G, C (Uracil instead of Thymine)
Nitrogen bases bonding	Bases paired — bonded to complementary strand	Bases overhung — NOT bonded to complementary strand
Location	Nucleus (Chromosomes)	Nucleolus, Cytoplasm
Function	Stores hereditary information	Helps in protein synthesis / translation

6.10 Function of RNA in Converting Hereditary Information into Useful Proteins

After synthesis in the nucleus, RNA is transported to cytoplasm. **RNA is an intermediate molecule for controlling the heredity**. The primary function of RNA is to **synthesize proteins by the process of translation**. All three types of RNA are involved in translation of genetic information into various proteins necessary for cellular processes.

6.10.1 Types and Functions of RNA

There are **three types** of RNA:

Type	Full Name	Function
mRNA	Messenger RNA	It carries message of DNA from nucleus to cytoplasm for protein synthesis, so called mRNA.
tRNA	Transfer RNA	It transfer amino acids to ribosomes during protein synthesis hence called tRNA.
rRNA	Ribosomal RNA	It is the component of ribosome therefore, called rRNA. It combines with ribosomal protein to form units of ribosomes, which are factory of proteins.

6.11 Flow of Genetic Information from DNA to RNA and then to Proteins

DNA (genes) contain information needed for the synthesis of proteins. The flow of biological information, from DNA to RNA and from RNA to protein is known as the **"central dogma of molecular biology"** or **"central dogma of life"**. Proteins are the main structural and functional molecules of living organisms. Correct Primary structure of any protein is primarily determined by DNA. Protein synthesis takes place in two sequential steps i.e. **transcription and translation**. Together transcription and translation are known as **gene expression** (Fig. 6.16).

■ Diagram: Central Dogma of Molecular Biology (Fig. 6.16)

```
┌──────────────┐  Transcription   ┌──────────┐  Translation   ┌──────────┐
│    DNA       │ ──────────────→ │   mRNA   │ ──────────→ │  Protein  │
│ (nucleus)    │                  │(cytoplasm)│              │          │
└──────────────┘                  └──────────┘              └──────────┘

DNA → (Transcription in Nucleus) → mRNA
mRNA → (Translation at Ribosome in Cytoplasm) → Protein

Together: Transcription + Translation = Gene Expression
```

6.11.1 Transcription

Definition: Transcription is the **first step of gene expression** that involves the **formation of RNA molecule from DNA**. Both DNA strands of a gene unwind from each other.

Process: Only one strand of DNA is copied. Nucleotide sequence of one DNA strand (coding strand) is copied in the form of **mRNA**. After the gene is copied to mRNA, both strands of DNA are again coiled together into its normal double stranded shape.

Location: Transcription takes place in the **NUCLEUS** of the cell.

Product: The product of transcription is **mRNA** which then moves from the nucleus to the cytoplasm.

6.11.2 Translation

Definition: Translation is the **second step of gene expression**.

The mRNA synthesized in the nucleus of the cell moves into the cytoplasm where it **binds to the ribosome**. The mRNA massage (codes) is translated by **ribosomes** which link amino acids to form **polypeptide chain with specific sequence of amino acids**. Different types of **tRNAs transfer amino acids to ribosome** as per sequence of codes in mRNA.

Location: Translation takes place in the **CYTOPLASM** at the **ribosomes**.

Product: A polypeptide chain / **protein** with a specific amino acid sequence.

Feature	Transcription	Translation
Process	DNA → mRNA (RNA synthesis)	mRNA → Protein (protein synthesis)
Location	Nucleus	Cytoplasm (at ribosomes)
Raw Materials	Ribonucleotides	Amino acids + tRNA + rRNA
Product	mRNA	Polypeptide / Protein

Chapter Summary — All Key Points (As in Textbook)

1.	Biochemistry is the study of different chemical compounds and the chemical processes taking place within the living organisms.
2.	Living organisms are composed of 16 out of 92 naturally occurring elements called bioelements out of which six are Major and ten are Minor bioelements.
3.	Different bio-elements combine to form inorganic and organic biomolecules including water, carbohydrates, lipids, proteins and nucleic acids.
4.	Proteins are made up of a single or more polypeptide chains which are polymers of amino acids. Amino acids are composed of C, H, O, N and sometimes P, S, Fe etc. Number, type and sequence of amino acids in each polypeptide chain is specific. Proteins play multiple structural and functional roles.
5.	A group of heterogeneous hydrophobic organic compounds including fats, oils, waxes and steroids found in the living organisms, is called lipids. One gram of lipids contain double amount of energy than carbohydrates due to higher carbon-hydrogen ratio.
6.	Most common types of lipids are tri-acyl glycerides which contains one glycerol and three fatty acids. Saturated fatty acids are solid at room temperature whereas unsaturated fatty acids are liquid at room temperature.

7.	Carbohydrates are the compounds of carbon, hydrogen and oxygen with a ratio of 1:2:1 respectively and literally called “hydrates of carbon”.
8.	Carbohydrates are the important biological molecules commonly called sugars or saccharides. Glucose is a monosaccharide, naturally produced in the green plants by the process of photosynthesis using CO_2 from air and H_2O from soil.
9.	Polysaccharides are complex carbohydrates of plants and animals to store extra carbohydrate e.g., starch, glycogen and are usually tasteless. Cellulose, chitin and peptidoglycan are polysaccharides which are structural components of cell walls in the plant, fungi and bacteria.
10.	Nucleic acids are of two types; Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA) which store and transfer hereditary information from one generation to next.
11.	DNA contain information for the synthesis of proteins. The flow of information, from DNA to RNA and from RNA to protein is known as the “central dogma of molecular biology” or “central dogma of life”. Together transcription and translation are known as gene expression.

Exercise Questions (From Textbook Pages 91–92)

Section I: Multiple Choice Questions

#	Question	Options (Answer underlined)
1	Which of the following is major bio-element?	A) sodium B) magnesium ✔C) Phosphorous D) iron
2	Peptide bond is formed between two:	A) Monosaccharides ✔B) Amino acids C) Nucleotides D) Fatty acids
3	During translation, sequence of amino acids in the protein is decides on the basis of sequence of nucleotides in:	A) mRNA B) rRNA C) tRNA ✔D) mRNA (mRNA codes)
4	Different amino acids differ from each other on the basis of their _____ group:	A) Alkyl B) Carboxylic C) Amino ✔D) varies by R group
5	Both strands of DNA are held together by hydrogen bonding, double hydrogen bonds are present between:	A) Adenine and thymine B) Cytosine and guanine C) Cytosine and thymine ✔D) Adenine and thymine
6	Transcription takes place in the:	A) Cytoplasm ✔B) Nucleus C) Ribosomes D) Rough Endoplasmic reticulum
7	All the nucleotides of RNA differ from the nucleotides of DNA in having different:	A) Nitrogen base ✔B) Pentose sugar C) Phosphate group D) carboxylic group
8	Diameter of DNA is thoroughly uniform and is about:	A) 34 nm B) 3.4 nm ✔C) 2nm D) 1.1nm
9	Which of the disaccharide is also called transport sugar?	A) Sucrose B) Maltose C) Fructose ✔D) Sucrose
10	Vitamin D belongs to:	A) Carbohydrates ✔B) Lipids C) Protein D) RNA

Section II: Short Answer Questions

#	Question
1	Compare ratios of bio-elements.
2	Compare the energy level of carbohydrates, proteins and lipids.
3	How biochemistry is important for study of physiology, cell biology and anatomy?
4	Define gene (a localized region of DNA that codes for a protein).
5	Name structural and energy producing nature of carbohydrates.

6	What are different plant sources of proteins?
7	How primary structure of protein is important?

Section III: Extensive Answer Questions

#	Question
1	How lipids are important for living organisms?
2	Explain the composition of chromatin material?
3	What are different types of disaccharides? How are they produced? Mentions their sources.
4	Give an account of genes on chromosomes.
5	Describe the role of gene in protein synthesis.
6	Explain how genes control inheritance of characters.
7	Describe the composition of chromatin material.
8	Explain that gene is a unit of inheritance and that it can be copied and passed on to the next generation.
9	Describe the central dogma stating the role of gene in protein synthesis.
10	Describe structure, sources and functions of lipids.

End of Chapter 6 — Molecular Biology Notes

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